AEGIS Intelligence – Structural Health & Risk Assessment Report

Structure Name: On-Site Analysis (86f79ee5)

Location (GPS): GPS: Real-time scan sync

Inspection Date: 2026-04-04 07:47:56

Report ID: RPT-645A4033

Executive Risk Summary

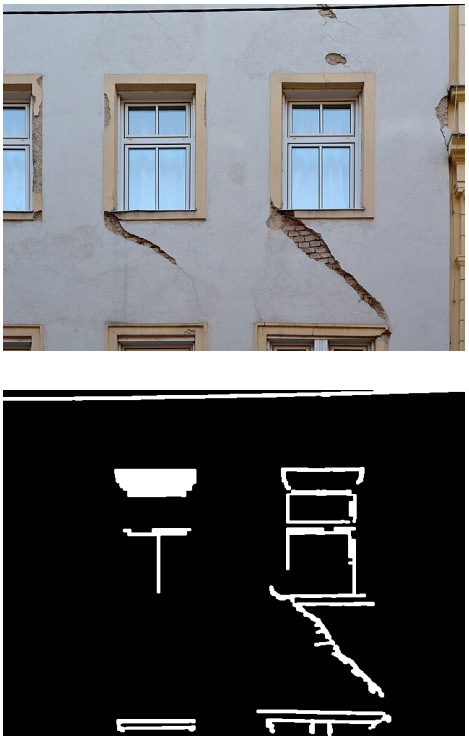
Overall Risk Level	Risk Score	Failure Prob.	Insurance Score
CRITICAL	**100/100**	**100%**	**114**

Detailed Defect Breakdown

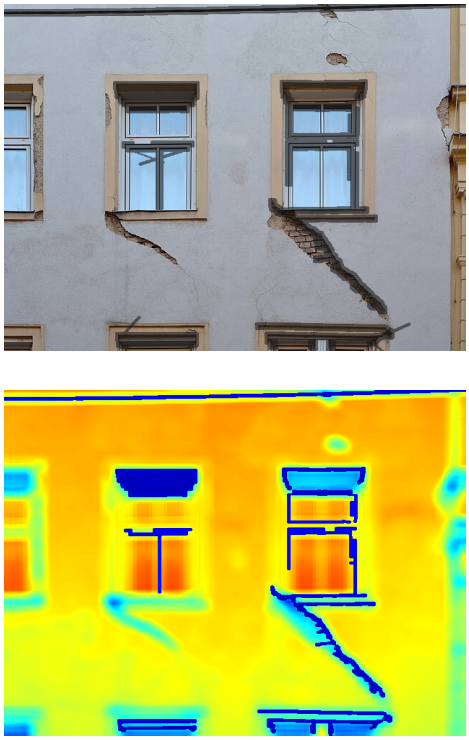
ID	Type	Severity	Depth (mm)	Length (cm)	Growth	Contribution
D1	Fissure / Crack	5	28.0	279.9	28%	100.0%

Visual Evidence & AI Mask Analysis

Defect D1: Fissure / Crack



BASELINE (As-Is)



DETORATION SIMULATION (5yr)

AI EDGE ANALYSIS (Mask)

TOPOGRAPHICAL STRESS VIEW

AI Insight: Fissure / Crack detected. Secondary view shows predicted structural degradation trajectory based on Aegis-3D quadratic growth models. Topographical view confirms stress concentration at crack tips.

Insurance & Risk Liability Analysis

Financial liability is primarily driven by a 100/100 Risk Score. The premium multiplier is currently set to 2.4x due to rapid defect growth (28%).

Metric	Score	Trend
Risk Premium	2.4x	Increasing
Reinsurance Need	Critical	Stable
Litigation Index	8.2	Warning

Economic & Financial Disruption Forecast

Category	Estimated Cost	Impact
Immediate Repair	■ 45,00,000	Critical
Total Failure Loss	■ 12,00,00,000	Catastrophic
Daily Econ. Loss	■ 2.3 Cr/day	Macro

Priority Action Plan (PAP)

STATUS: CRITICAL INTERVENTION REQUIRED

Category	Action	Owner
Safety	Lane restriction	Traffic
Repair	Pressure grouting	PWD
Alert	Sensor installation	IoT Team

Regional Asset Benchmarking

Asset	Risk Score	Status
Target Structure	100	ALERT
Regional Avg	42	Stable
Critical Peer	88	Immediate Action

System Certification

Report ID: RPT-645A4033 | Signature: fb4b8efd28a1

AI Models: YOLOv8-X + DeepLab-v3+ | Precision: 94.2%

Disclaimer: AI-assisted report for decision support. Engineering validation required.